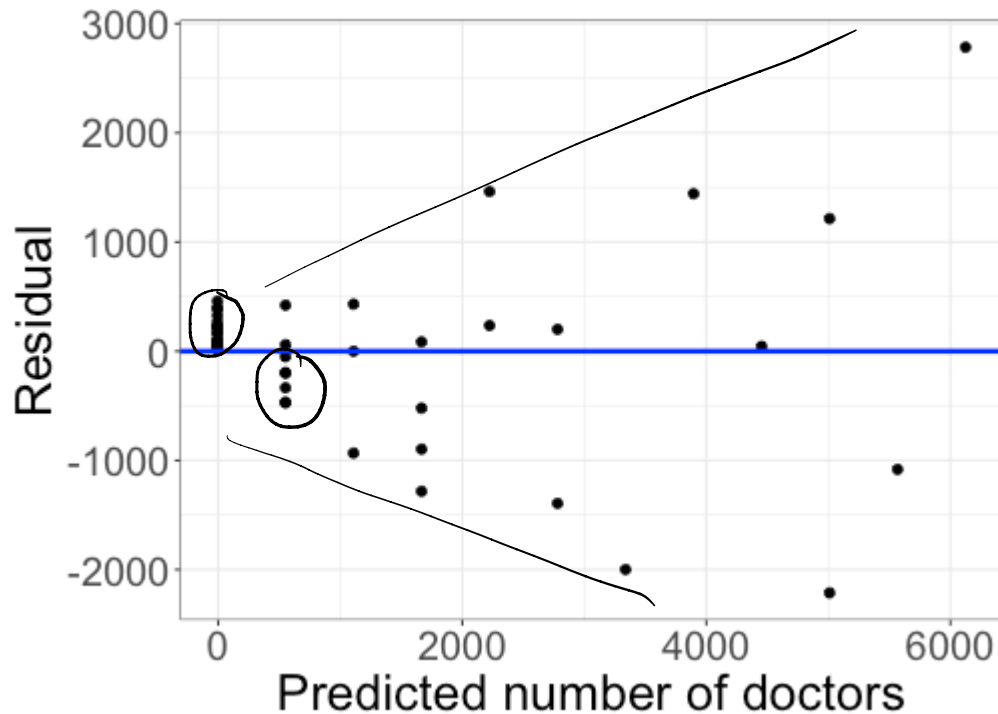


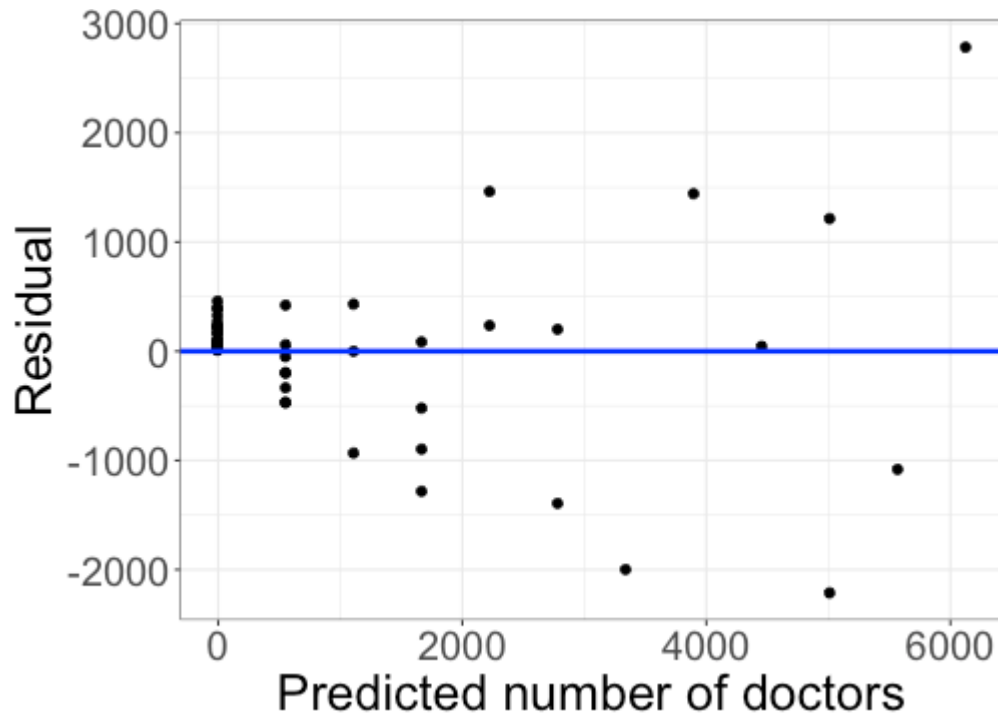
Nonlinear relationships and transformations

Last time



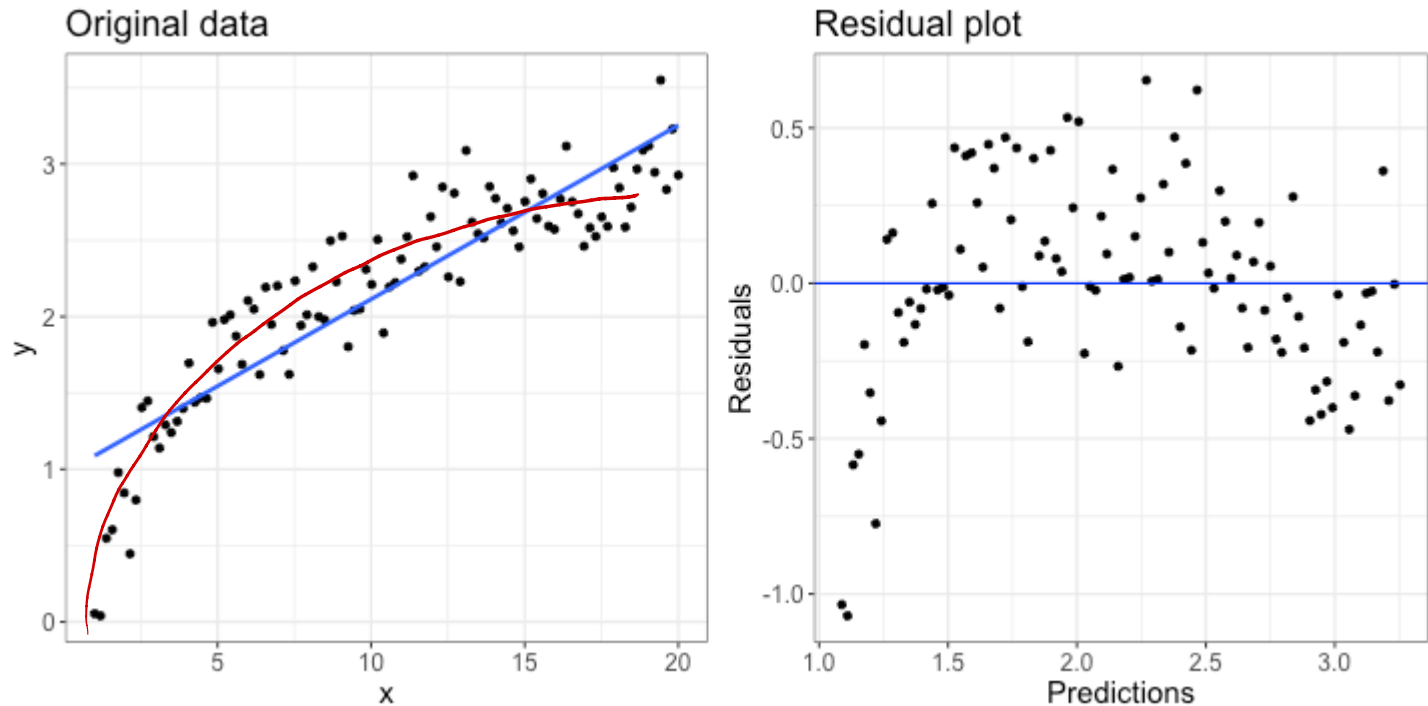
Do the shape and constant variance assumptions look reasonable?

Today



Today: handling non-linear relationships

Example

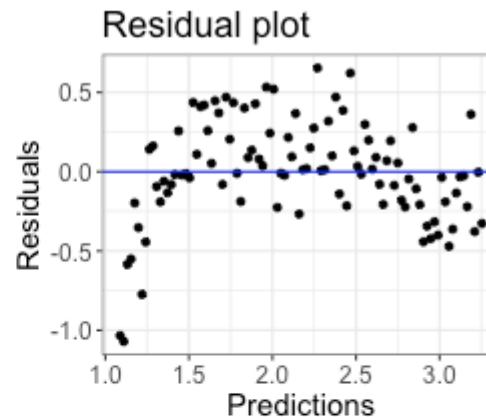
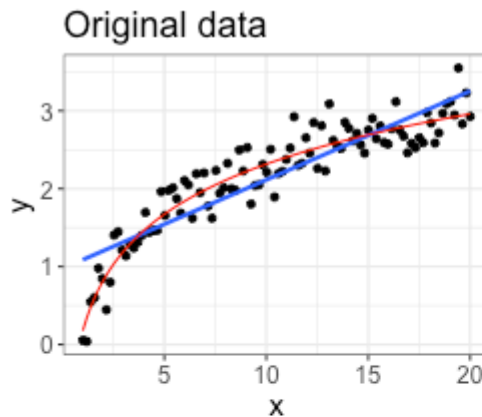


The relationship between x and y does not appear linear. How could we model that relationship instead?

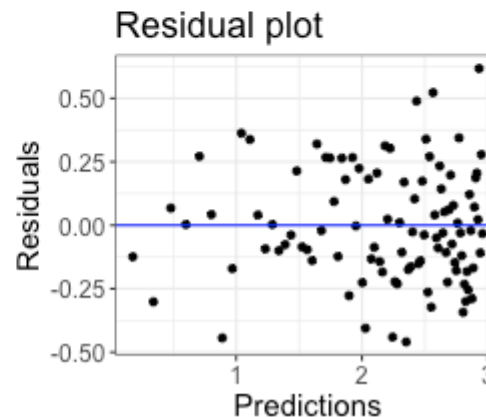
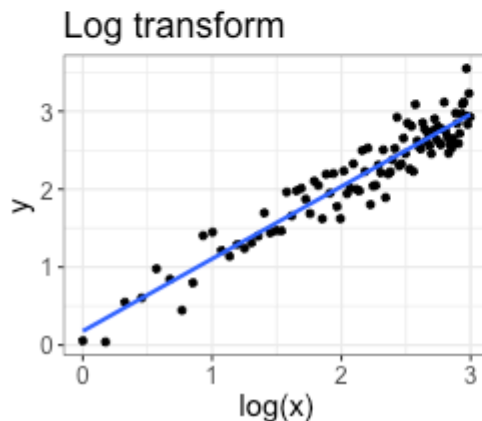
Example: log transform

$$Y = \beta_0 + \beta_1 \log(X) + \varepsilon$$

↙ natural log (base e)
↘ not linear in x ,
but is linear in $\log(x)$

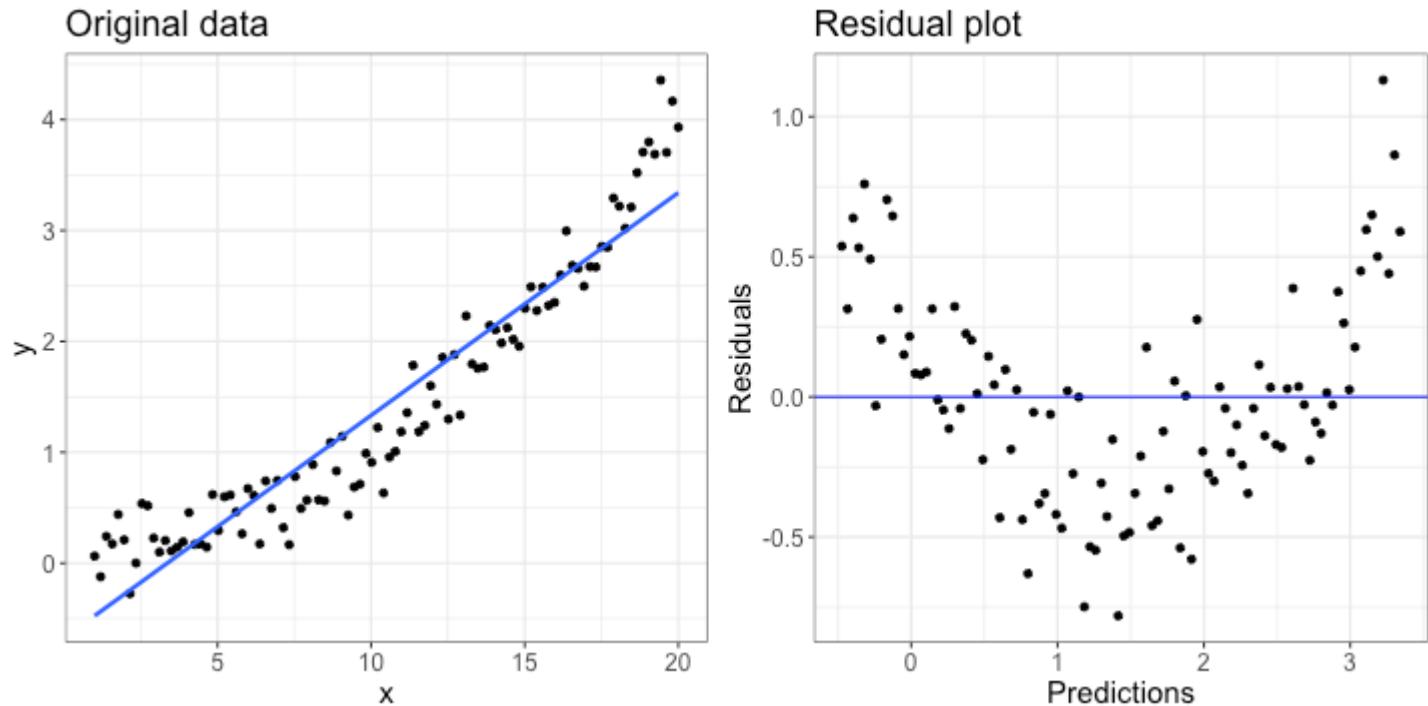


↙ shape assumption
not satisfied



↙ shape
assumption
is satisfied

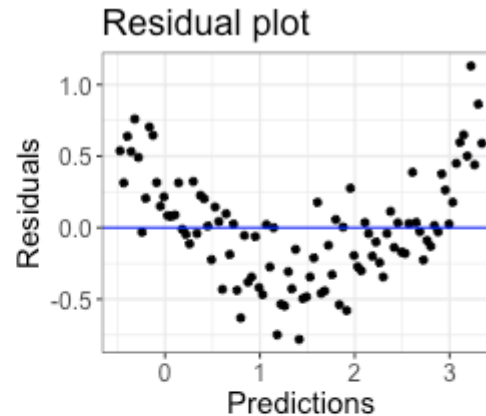
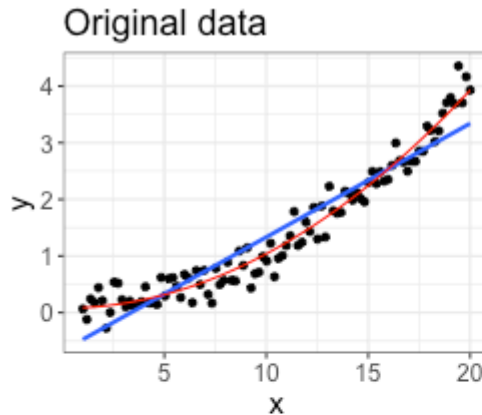
Example



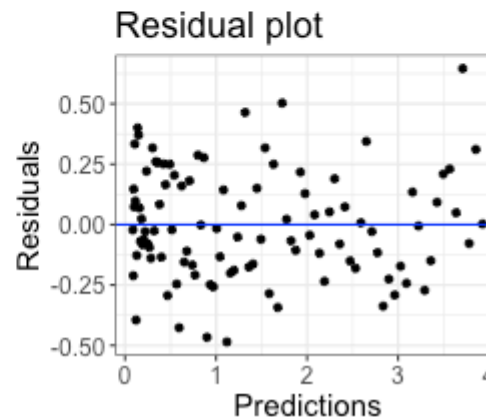
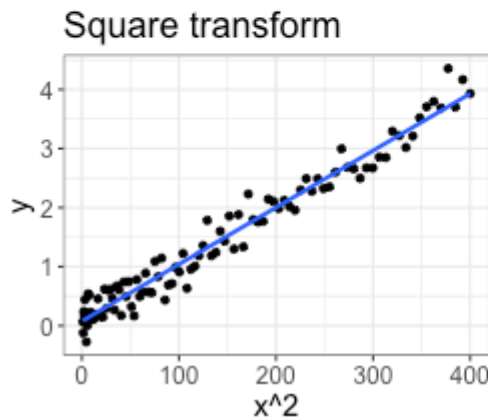
The relationship between x and y does not appear linear. How could we model that relationship instead?

Example: squaring X

$$Y = \beta_0 + \beta_1 X^2 + \varepsilon$$



← shape assumption is not satisfied

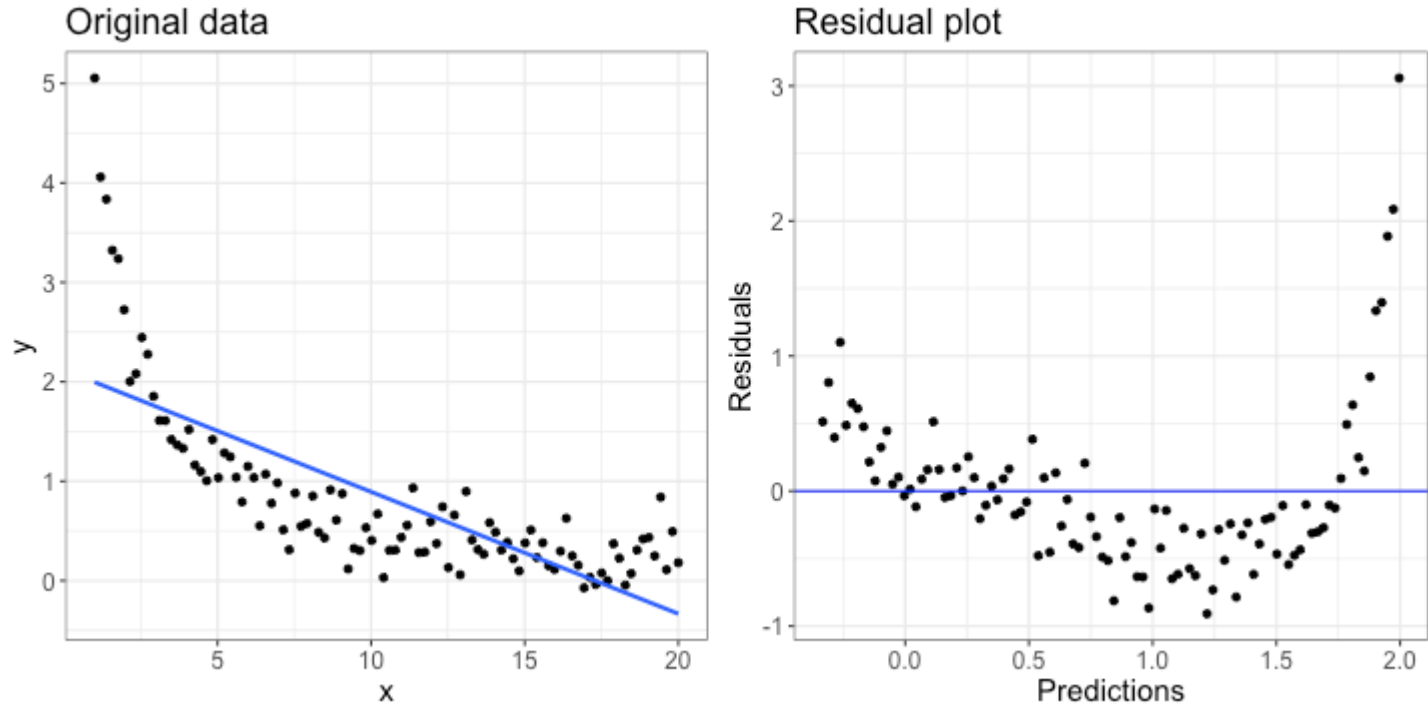


← shape assumption looks better!

$\frac{1}{x}$



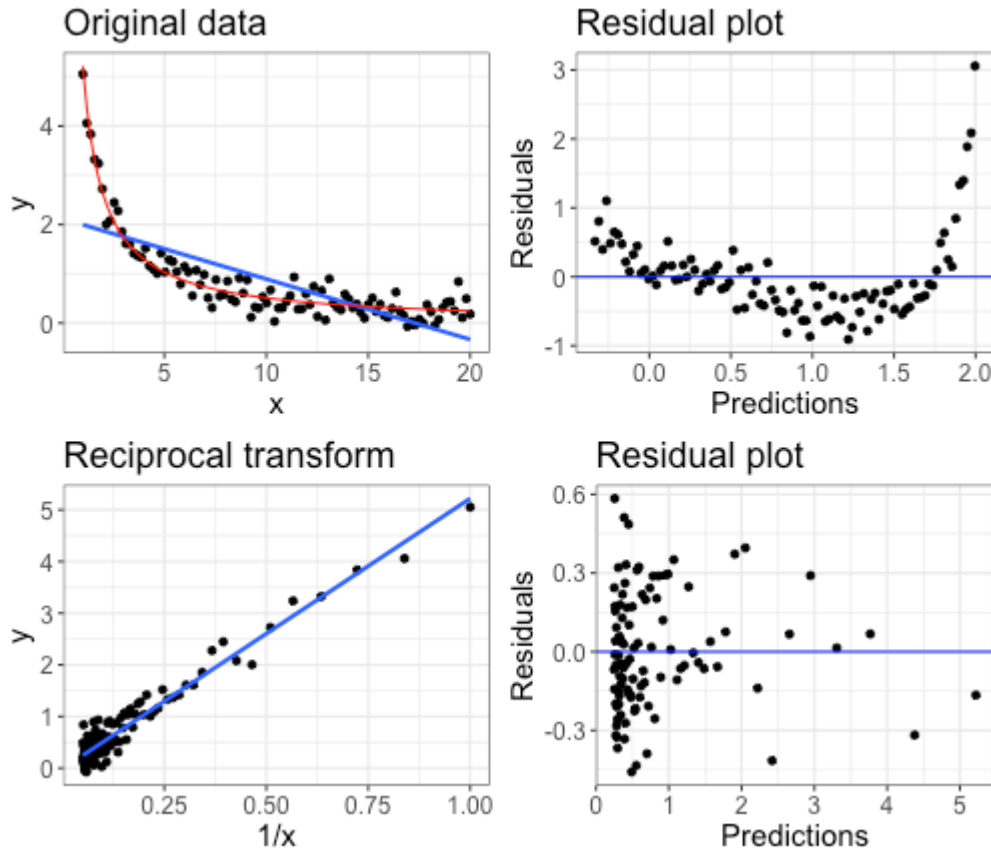
Example



The relationship between x and y does not appear linear. How could we model that relationship instead?

Example: reciprocal transform

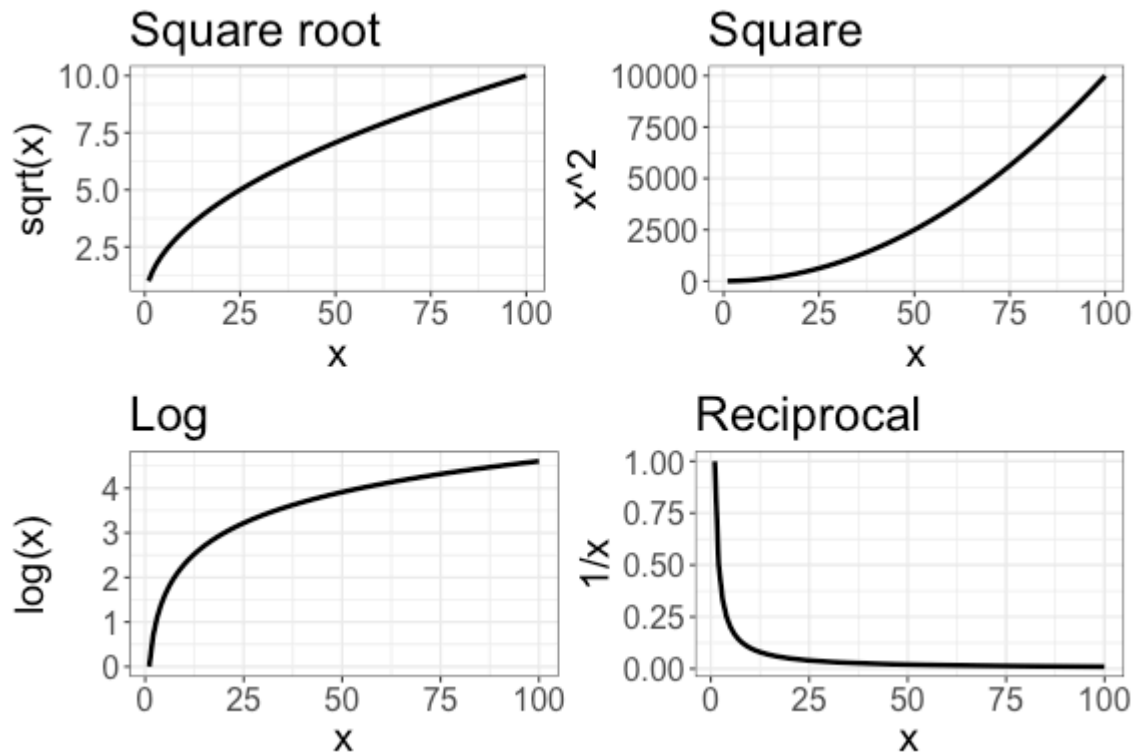
$$Y = \beta_0 + \beta_1 (1/X) + \varepsilon$$



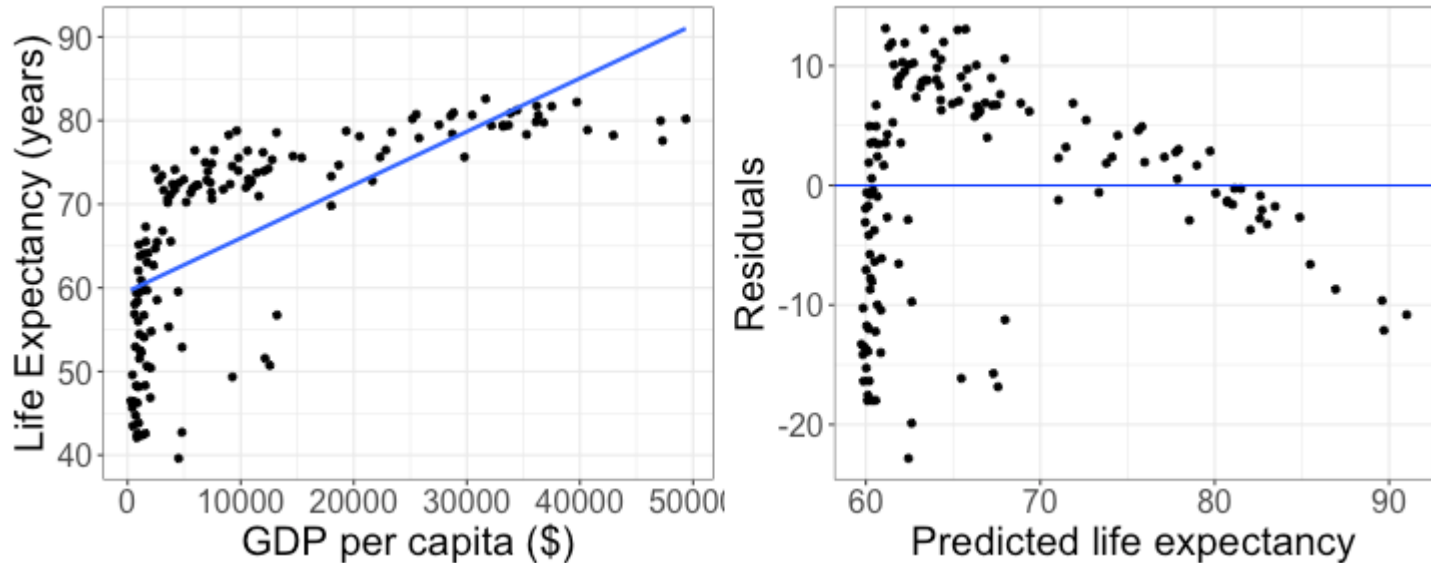
← Shape assumption looks better!

Choosing transformations

To model nonlinear relationships, try transformations for x that looks like the relationship

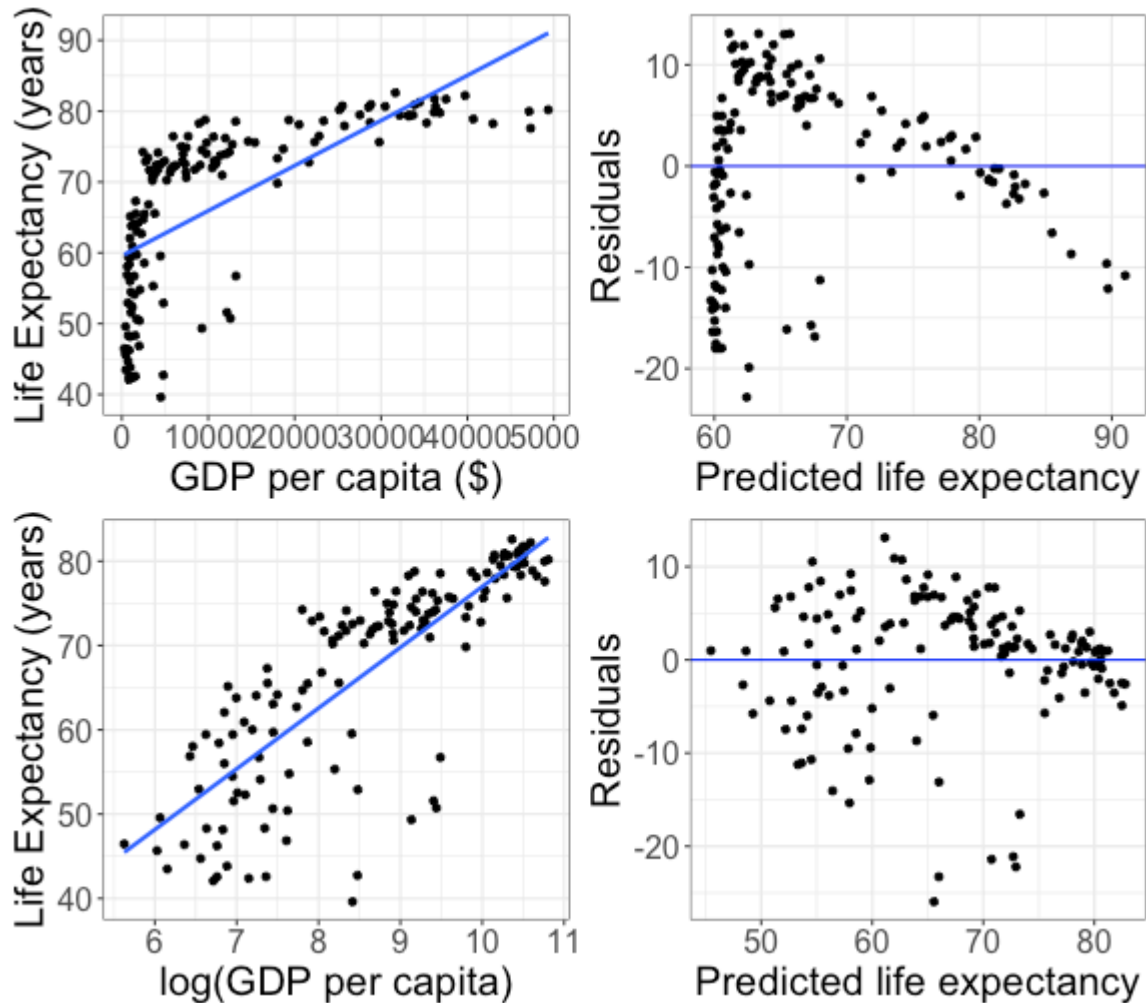


Example

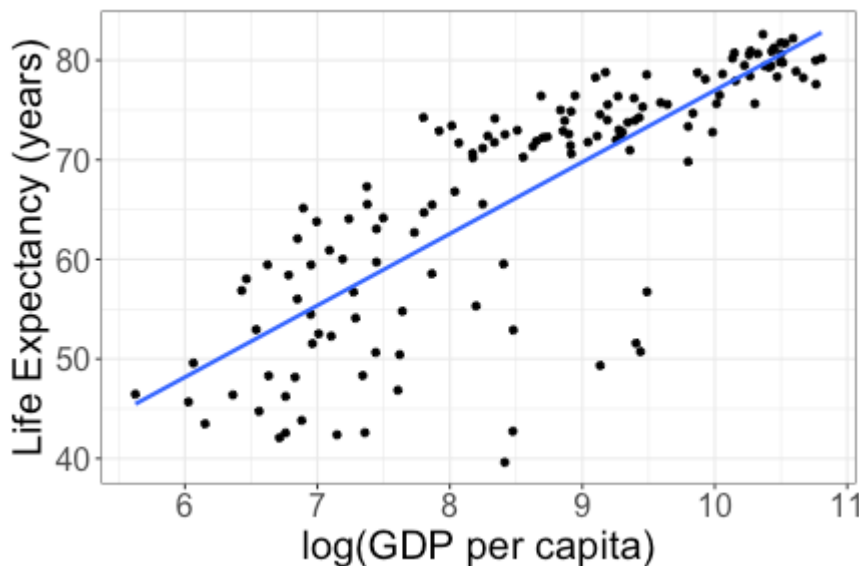


What transformation could I try for this data?

Log transformation



Log transformation

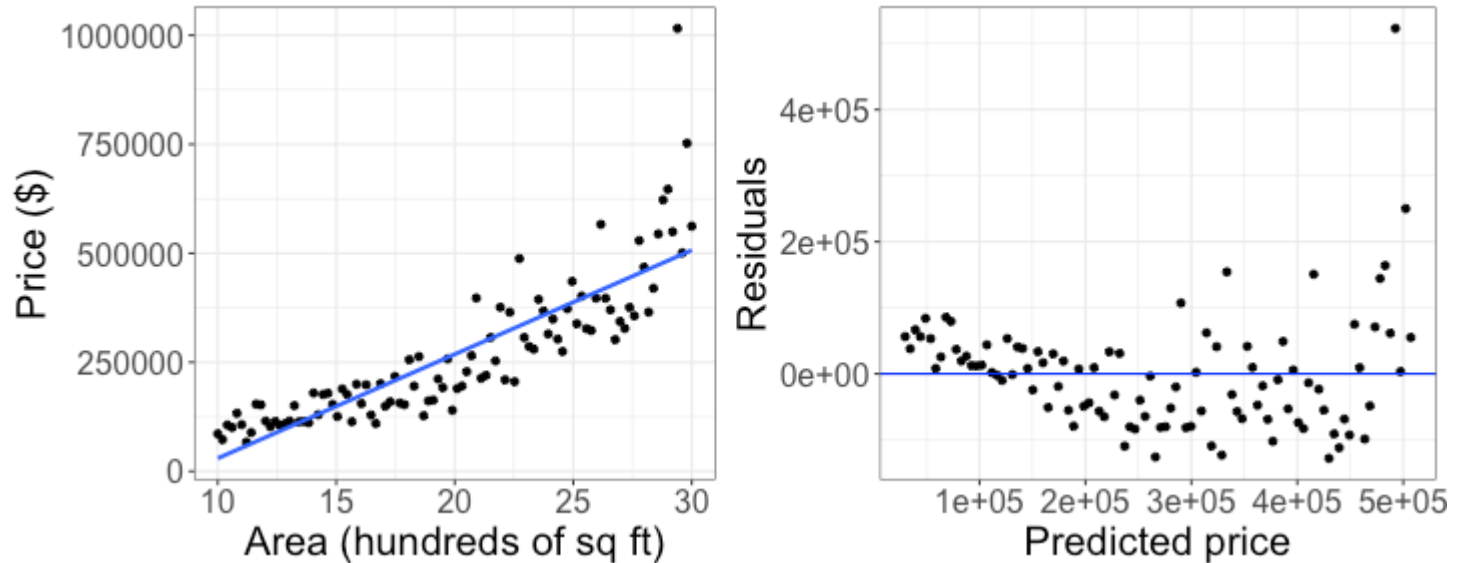


$$\widehat{\text{Life Expectancy}} = 4.95 + 7.20 \log(\text{GDP per capita})$$

What life expectancy would I predict for a GDP per capita of \$20000?

$$\begin{aligned} & 4.95 + 7.20 \log(20000) \\ &= 4.95 + 7.2(9.9) = 76.26 \text{ (years)} \end{aligned}$$

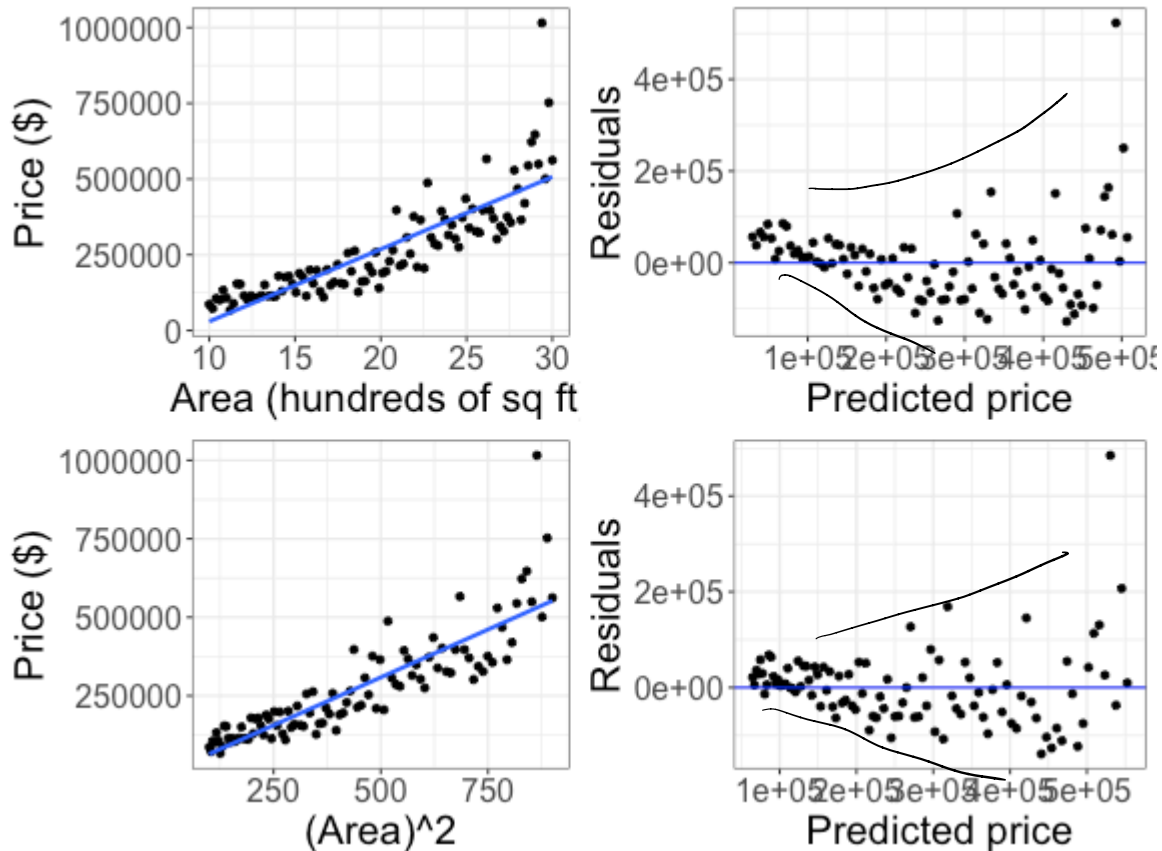
Example



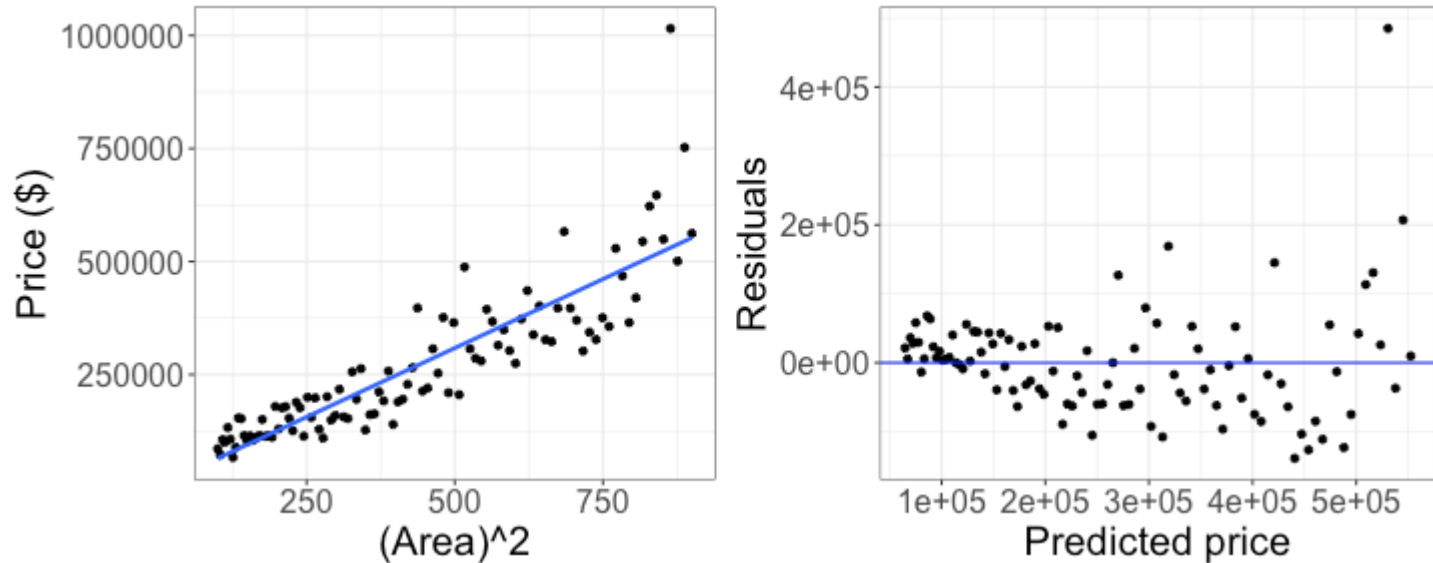
What transformation could I try for this data?

Trying squaring X

$$\text{Model: } Price = \beta_0 + \beta_1 (Area)^2 + \varepsilon$$



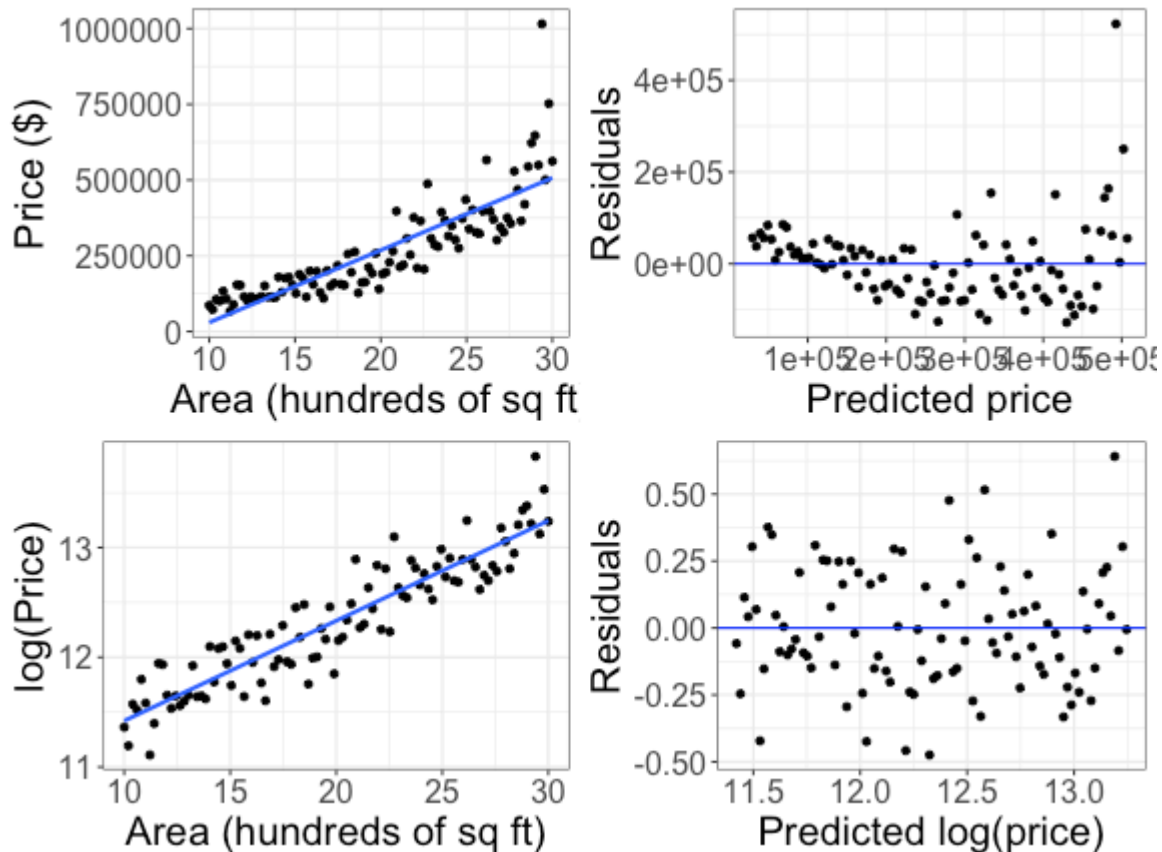
Trying squaring X

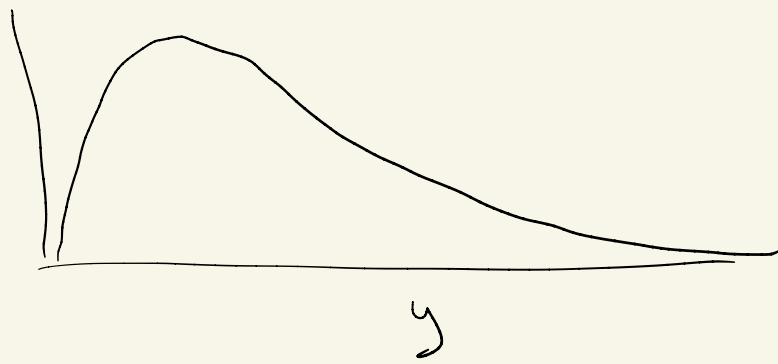


What do you notice about the residual plot after trying the transformation?

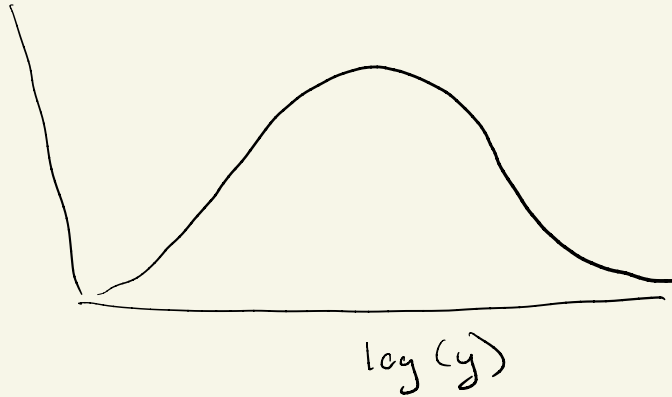
Another attempt: transforming the response!

$$\text{Model: } \log(\text{Price}) = \beta_0 + \beta_1 \text{Area} + \varepsilon$$





(right-skewed variable)



(less right-skewed)

$\log(-1)$ or $\sqrt{y}$ can help address non-constant variance

Log transformed response

$$\widehat{\log(\text{Price})} = 10.51 + 0.091 \text{ Area}$$

+ Here Price is in \$, Area is in hundreds of square feet

How would I calculate the predicted *price* (in \$) for a house of 2000 square feet?

$$\begin{aligned} 2000 \text{ ft}^2 &= 20 \times 100 \text{ ft}^2 \\ \Rightarrow \text{Area} &= 20 \end{aligned}$$

$$10.51 + 0.091(20) = 12.33$$

$$\widehat{\log(\text{Price})} = 12.33$$

$$\Rightarrow \widehat{\text{Price}} = e^{12.33} = \$226,386$$

Class activity

Work on the handout, then we will discuss as a class

Class activity

$$\widehat{\text{Life Expectancy}} = 4.95 + 7.20 \log(\text{GDP per capita})$$

life expectancy
What is the predicted ~~price~~ for GDP per capita of \$20200?

$$4.95 + 7.20 \log(20200) = 76.327$$

Class activity

$$\widehat{\text{LifeExpectancy}} = 4.95 + 7.20 \log(\text{GDP per capita})$$

+ When GDP per capita = 20000: $\widehat{\text{LifeExpectancy}} = 76.255$

+ When GDP per capita = 20200: $\widehat{\text{LifeExpectancy}} = 76.327$

Change:

$$76,327 - 76,255 = 0.072$$

$$= \frac{7.2}{100}$$

Interpretation: a 1% increase in GDP per capita is associated with an increase of 0.072 years in life expectancy, on average

Class activity

$$\widehat{\log(\text{Price})} = 10.51 + 0.091 \text{ Area}$$

+ Price in \$, Area in hundreds of sq ft

$\Rightarrow \text{Area} = 21$
(hundreds of ft²)

What is the predicted price for a house of 2100 square feet?

$$\begin{aligned} & 10.51 + 0.091(21) \\ & = 12.421 \\ \hat{\text{Price}} & = e^{12.421} = \$247,954 \end{aligned}$$

Class activity

$$\widehat{\log(\text{Price})} = 10.51 + 0.091 \text{ Area}$$

- + Price in \$, Area in hundreds of sq ft
- + When Area = 20: $\widehat{\text{Price}} = 226386$
- + When Area = 21: $\widehat{\text{Price}} = 247954$

Change:

$$\frac{247954}{226386} = 1.095 = e^{0.091}$$

Interpretation:

a one-unit increase in Area
(i.e. an increase of 100 ft²) is
associated with an increase in price
by a factor of $e^{0.091} = 1.095$,
on average

Class activity

$$\log(\widehat{\text{Price}}) = 10.51 + 0.091 \text{ Area}$$

- + Price in \$, Area in hundreds of sq ft
- + When Area = 20: $\widehat{\text{Price}} = 226386$
- + When Area = 21: $\widehat{\text{Price}} = 247954$

$$\text{Change: } \frac{247954}{226386} = 1.095 = e^{0.091}$$